MACROECONOMIC FORECASTING AND INFLATION SURGE DETECTION

A Multi-Indicator Machine Learning and Time-Series Framework Using FRED Data

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Research report and experimental documentation

Study focus: Semiconductor and Other Electronic Component Manufacturing Producer Price Index (PPI)

Forecast origin used in the final integrated run: September 2026

Forecast horizon: October 2026 – March 2027

Prepared from the uploaded Colab notebook, experimental outputs, and forecast visualizations supplied in this conversation.

Abstract

This study develops a monthly macroeconomic forecasting framework for semiconductor-sector producer price inflation using six FRED indicators: semiconductor manufacturing PPI, copper prices, aluminum prices, nickel prices, semiconductor capacity utilization, and semiconductor industrial production. The workflow combines exploratory data analysis, feature engineering, chronological forecasting evaluation, ARIMA/SARIMA/SARIMAX models, rolling-window estimation, residual diagnostics, prediction-interval analysis, inflation-surge labeling, and direct machine-learning forecasts at horizons one through six months. The analysis identifies substantial distribution change and a sharp 2026 inflation acceleration. A rolling 180-month SARIMAX specification using semiconductor industrial production is the strongest statistical specification in the reported six-month-ahead comparison, while a separate direct-forecast experiment selects models independently for H1–H6. The final integrated run produces six direct PPI-inflation forecasts from a September 2026 origin. Raw-material forecasts and a smartphone cost-pass-through illustration are included as supplementary scenario outputs, not validated consumer-price predictions.

Executive summary

Research question. Can historical semiconductor manufacturing conditions and selected commodity indicators improve forecasts of semiconductor-sector PPI inflation, and can unusually large inflation accelerations be flagged using only historical information?

Main evidence. The 180-month rolling SARIMAX model with semiconductor industrial production achieved MAE 1.6989, RMSE 2.6923 and R^2 0.2960 over 73 six-month-ahead evaluation observations. It outperformed the expanding-window and rolling-120-month alternatives in the reported comparison. The direct H1–H6 validation experiment selected Random Forest for H1/H3, XGBoost for H2/H5/H6, and persistence for H4.

Important qualification. The final six-month forecast and the historical six-month-ahead SARIMAX results are different experiments with different target constructions and evaluation windows. They must be reported separately. The phone-price output is an illustrative cost-pass-through scenario based on assumed BOM shares; it is not a trained phone retail-price forecast.

Key reported outputs

Item	Reported result
Master dataset	417 monthly rows; Jan 1992–Sep 2026; 9 columns
Direct-horizon dataset	393 origins; Jul 1993–Mar 2026; 40 predictors; H1–H6 targets
Best statistical specification	Rolling 180-month SARIMAX with industrial production
SARIMAX six-month-ahead MAE	1.6989 percentage points (73 observations)
Surge threshold	2.3223 pp in 3-month YoY inflation momentum
Detected surge months	15 months; 3.82% of 393 observations
Final integrated forecast origin	Sep 2026
Final forecast period	Oct 2026–Mar 2027

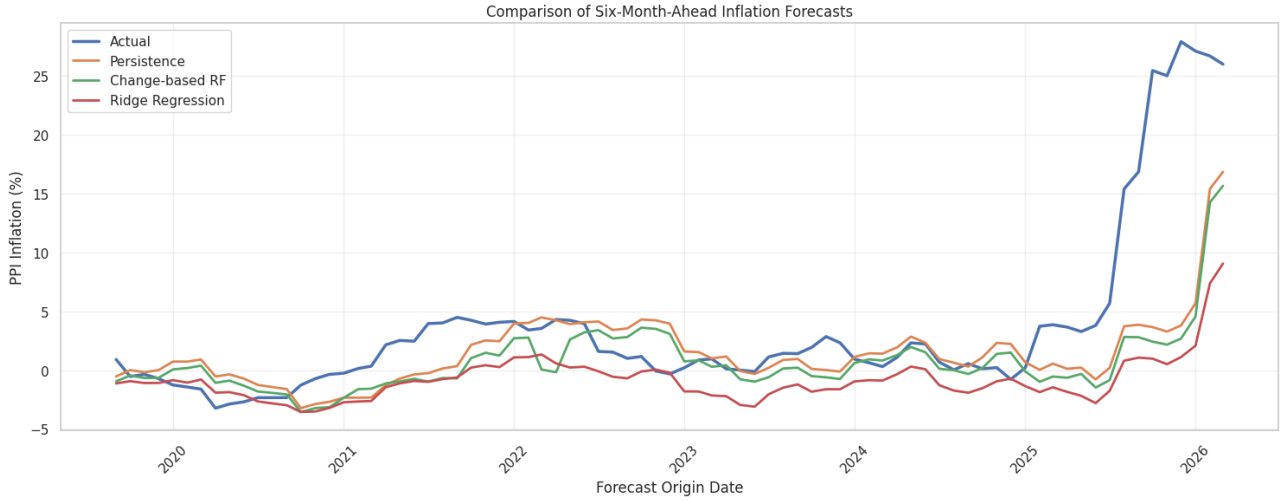


Figure 1. Historical trends in the six FRED indicators. Use this figure to establish the long-run behaviour and scale differences across the input series.

1. Research scope and interpretation

The target variable is year-over-year producer price inflation for the FRED industry series PCU3344133441, described as Producer Price Index by Industry: Semiconductor and Other Electronic Component Manufacturing. The project forecasts sector-level producer-price inflation. It does not directly forecast the market price of semiconductor chips, finished electronic devices, or smartphones.

The working project title refers to a multi-indicator framework inspired by macroeconomic inflation forecasting. The analysis does not reproduce a proprietary Goldman Sachs forecasting pipeline, and no claim is made that Goldman Sachs uses the same model, features, or parameters.

The smartphone component is included only as a downstream illustration. A genuine smartphone retail-price model would require historical handset prices, bill-of-materials data, display/memory/battery/camera costs, assembly and logistics costs, exchange rates, duties/taxes, inventory and retailer/brand margins.

2. Data and variables

The master dataset contains 417 monthly observations from January 1992 through September 2026. The six underlying FRED series are:

FRED series	Variable	Role
PCU3344133441	Semiconductor and other electronic component manufacturing PPI	Target level; YoY inflation derived
PCOPPUSDM	Copper price (USD per metric ton)	Commodity indicator
PALUMUSDM	Aluminum price (USD per metric ton)	Commodity indicator
PNICKUSDM	Nickel price (USD per metric ton)	Commodity indicator
CAPUTLG3344S	Semiconductor manufacturing capacity utilization	Manufacturing indicator
IPG3344S	Semiconductor industrial production	Manufacturing indicator

The notebook reports 12 missing values in each of the two derived PPI target columns, 24 missing cells in total, and no duplicate records. The model-ready dataset after dropping unavailable target rows contained 399 observations. The direct-horizon target builder subsequently produced 393 rows from July 1993 through March 2026, because each forecast horizon requires an observed future target.

Feature engineering included level and change transformations, lagged inflation, commodity changes, manufacturing changes, and horizon-specific targets. The final direct-forecast experiment used 40 numeric predictors before later target exclusion/feature filtering.

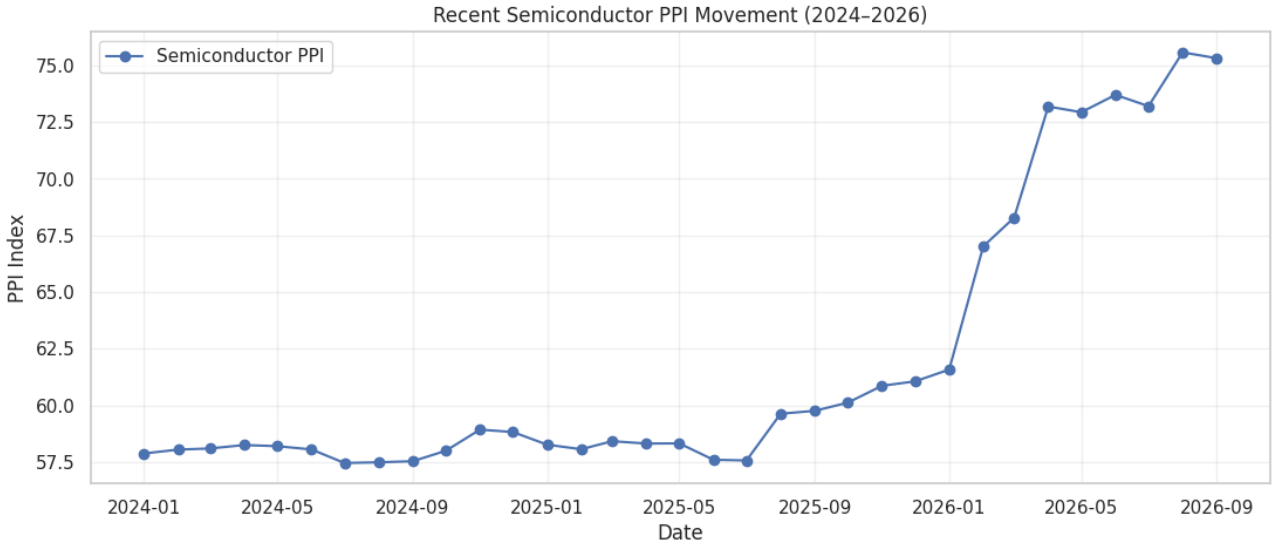


Figure 2. Semiconductor manufacturing PPI level, 1992–2026. This is a central descriptive figure for the paper.

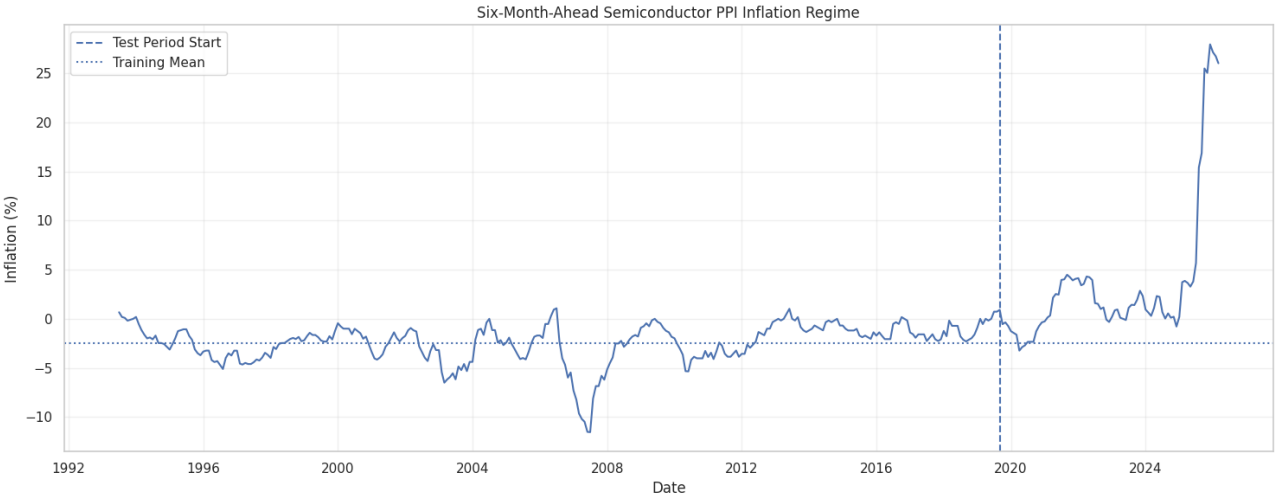


Figure 3. Correlation matrix across the macroeconomic and commodity indicators. Include as exploratory evidence, not as a causal result.

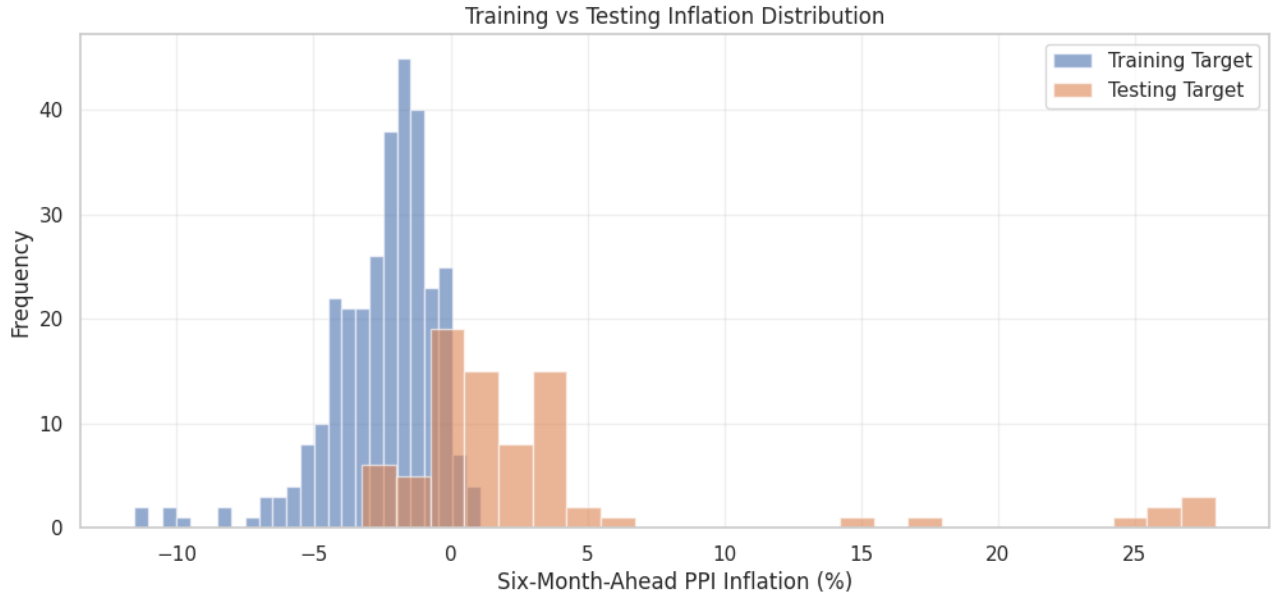


Figure 4. Distribution diagnostics. This figure is useful for explaining skewness, tails and the presence of extreme observations.

3. Exploratory data analysis

The notebook examines data quality, descriptive statistics, individual time-series trends, distribution shape, outliers, Pearson/Spearman correlation, and stationarity. These steps establish the time-series properties that motivate differencing, lag construction, and statistical-model diagnostics.

3.1 PPI level and inflation dynamics

The semiconductor PPI level is relatively subdued for much of the historical sample, followed by a sharp increase in 2026. The notebook specifically flags the 2026 movement for verification because it is unusually large compared with the earlier history. This behaviour creates a potential structural change and a difficult out-of-distribution forecasting environment.

The derived YoY inflation series exhibits substantial variation across historical regimes. In the later data supplied in the conversation, reported values include 5.6820% in January 2026, 15.4107% in February 2026, and 16.8712% in March 2026; the integrated runtime subsequently reports a September 2026 forecast origin with latest historical YoY inflation around 26.0152%. These are vintage-dependent values and should be reconciled against the final downloaded FRED vintage before publication.

3.2 Correlation and feature relationships

Correlation analysis is descriptive, not causal. Correlation between commodity prices, manufacturing activity and PPI inflation does not demonstrate that one variable causes another. The notebook includes both Pearson and Spearman analyses and visual correlation matrices.

3.3 Distribution and outlier behaviour

Histogram and boxplot analysis are used to assess skewness, heavy tails and outliers. Such properties are especially relevant because the later residual diagnostics show extreme non-normality. The target distribution should not be assumed Gaussian.

4. Forecasting methodology

4.1 Target definition and forecast horizons

Let y_t denote the semiconductor manufacturing PPI YoY inflation rate at month t. The earlier six-month-ahead experiment forecasts y_(t+6). The later direct-horizon experiment constructs six separate targets:

Target_Hh(t) = y_(t+h), for h = 1, 2, 3, 4, 5, 6.

Each horizon is modelled separately using information available at the forecast origin t. This is a direct multi-horizon strategy rather than recursively feeding one forecast into the next. The direct-horizon dataset contains 392, 391, 390, 389, 388 and 387 valid samples for H1 through H6, respectively.

4.2 Chronological evaluation

For the direct-horizon experiment, the reported split uses training origins July 1993–December 2016 (282 observations), validation origins July 2017–December 2019 (30 observations), with a six-month gap between them. Development observations begin July 2020. The split is chronological to reduce look-ahead leakage and to preserve the temporal order of observations.

For the earlier six-month-ahead statistical/ML experiments, the notebook reports a chronological evaluation period of September 2019–March 2026 with 73 forecast observations in the SARIMA/SARIMAX comparisons. Results from this experiment should not be merged numerically with the later 30-observation direct-horizon validation results.

4.3 Baseline and machine-learning models

Persistence provides a transparent benchmark: the forecast at horizon h is set equal to the latest observed inflation value at the origin. Ridge regression provides a regularized linear benchmark. Random Forest captures nonlinear interactions through an ensemble of decision trees. XGBoost provides a gradient-boosted tree benchmark. The direct-horizon experiment selects the lowest-validation-MAE model separately for each horizon.

4.4 Statistical time-series models

ARIMA models temporal dependence after differencing. SARIMA adds seasonal terms with a 12-month period for monthly data. SARIMAX adds external indicators. In the notebook's SARIMAX forecasting strategy, future exogenous variables are not observed at the forecast origin and therefore require a forecasting or scenario assumption; the experiment uses a persistence assumption for future industrial-production exogenous values.

4.5 Rolling and expanding windows

An expanding window retains all available historical observations. A rolling window uses a fixed recent history, allowing older regimes to leave the estimation sample. The notebook compares 120-month and 180-month rolling windows against an expanding window to assess sensitivity to structural change.

4.6 Evaluation metrics

MAE = (1/n) Σ |y_t – ■_t|. RMSE = sqrt[(1/n) Σ (y_t – ■_t)²]. R^2 measures variance explained relative to the evaluation-sample mean. Bias is the mean signed forecast error. MAE and RMSE are expressed in percentage points of YoY inflation. Lower MAE/RMSE is better; R^2 can be negative when the model performs worse than the evaluation-sample mean benchmark.

5. Results: earlier six-month-ahead forecasting

5.1 Machine-learning and baseline results

The notebook initially finds that complex models do not automatically outperform simple benchmarks. The persistence forecast is competitive and at several stages outperforms conventional Random Forest, XGBoost and Ridge models. A change-based Random Forest experiment improves on level prediction, illustrating the importance of target formulation.

Model / experiment	MAE	RMSE	R^2
Dummy baseline	5.9509	9.3683	-0.6625
Random Forest (all features)	5.1523	8.8063	-0.4691
XGBoost	4.9023	8.6866	-0.4294
Ridge (inflation history)	4.4285	7.4066	-0.0392
Change-based Random Forest	3.5870	6.4331	0.2160
Persistence	3.3251	6.0287	0.3115
ARIMA(1,1,1)	3.2389	5.8719	0.3468
Expanding SARIMA	3.1040	5.6197	0.4017
Rolling SARIMA (120 months)	2.9637	5.6766	0.3896

These results belong to the earlier shifted six-month-ahead target experiment. They are exploratory/development results because the same 2019–2026 period was repeatedly examined during model development.

5.2 SARIMA and rolling-window comparison

The notebook reports that SARIMA outperformed ARIMA and persistence in most historical validation folds. Rolling SARIMA reduced MAE relative to expanding SARIMA, while the expanding model sometimes retained better RMSE or R^2. This illustrates the trade-off between adapting to recent regimes and retaining a longer historical sample.

5.3 SARIMAX and indicator ablation

Model / exogenous setup	MAE	RMSE	R^2
SARIMA baseline	2.0993	3.1258	0.0510
SARIMAX, five indicators	1.9982	2.9764	0.1396
SARIMAX, manufacturing indicators	1.9304	2.9020	0.1821
SARIMAX, industrial production only	1.9260	2.9084	0.1785

The ablation suggests that semiconductor industrial production contributes more predictive value in this experiment than the commodity indicators considered individually. The five-indicator SARIMAX does not outperform the manufacturing-only variant. These are predictive associations, not causal effects.

5.4 Rolling 180-month SARIMAX result

Window strategy	N	MAE	RMSE	R^2	Bias
Rolling 180 months	73	1.6989	2.6923	0.2960	0.8639
Rolling 120 months	73	1.8060	2.7494	0.2658	0.9746
Expanding window	73	1.9260	2.9084	0.1785	1.1843

Approximate paired HAC comparisons reported in the notebook indicate an MAE improvement of 0.2271 for rolling 180 months versus expanding (approximate $p = 0.0122$), and 0.1071 versus rolling 120 months (approximate $p = 0.0355$). These should be described cautiously: forecast errors overlap across six-month horizons, and repeated development on the same period limits confirmatory interpretation.

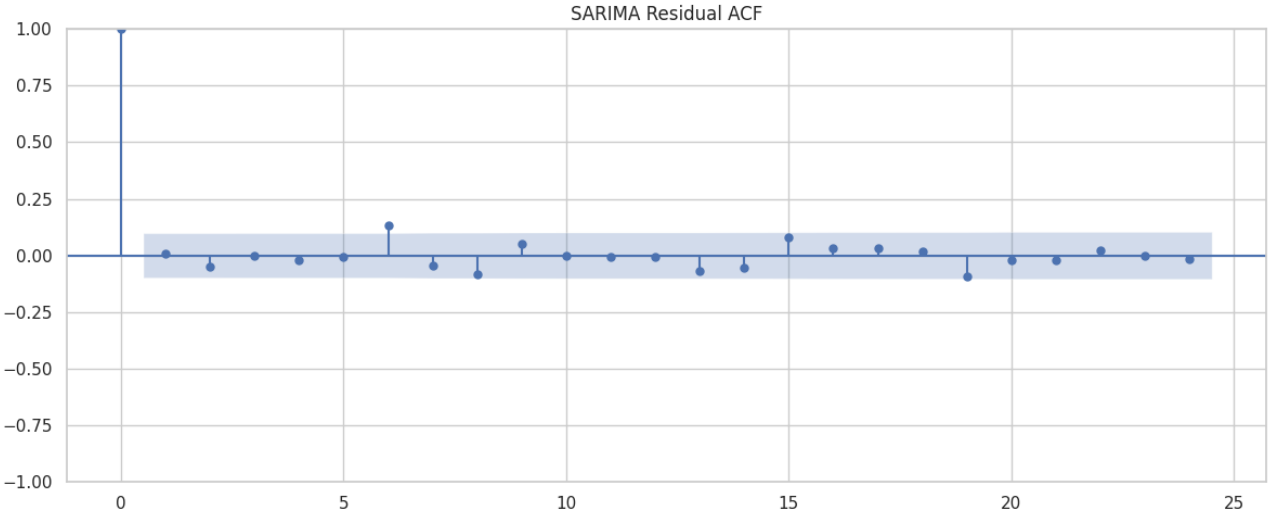


Figure 5. Residual diagnostic panel for the rolling 180-month SARIMAX model. Useful to report autocorrelation and heavy-tail limitations.

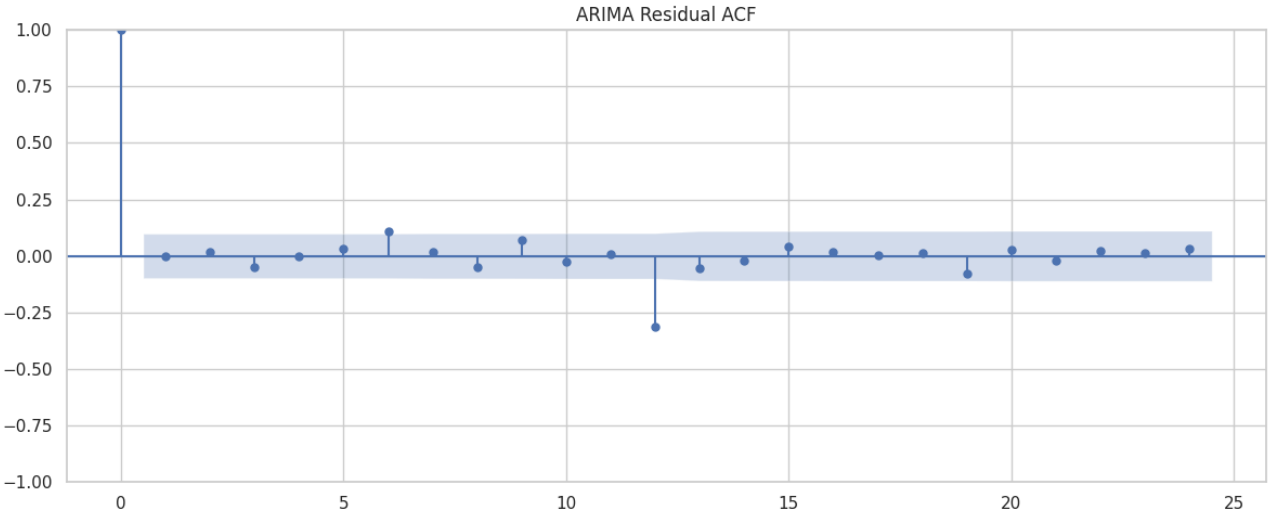


Figure 6. Prediction interval / calibration diagnostic from the notebook. Include to support the uncertainty discussion.

6. Diagnostics, uncertainty and regime analysis

6.1 Residual diagnostics

For the fitted rolling 180-month SARIMAX model, the notebook reports 180 residual observations, convergence, residual mean 0.0502, standard deviation 0.9893, skewness 3.3961 and kurtosis 36.6463. The Jarque–Bera statistic is 10418.1468 with p-value approximately zero, indicating strong non-normality. Ljung–Box p-values are 0.2002 at lag 12 and 0.5540 at lag 24, so the tests do not detect significant residual autocorrelation at those tested lags.

6.2 Inflation regime performance

Regime	N	SARIMA MAE	SARIMAX MAE	MAE change
High inflation (>5%)	3	11.0588	9.8241	11.17% improvement
Low inflation (<0%)	20	1.1290	1.3272	17.56% worse
Moderate inflation (0–5%)	50	1.9499	1.3600	30.25% improvement

The high-inflation regime contains only three observations, so its comparison is highly uncertain. The industrial-production SARIMAX improves errors in the moderate regime but performs worse in the low-inflation regime. The model still underpredicts the sharp 2026 acceleration.

6.3 Prediction intervals and conformal calibration

Interval method	Nominal	Observed coverage	Average width
Gaussian / model-based	80%	71.23%	3.3374
Gaussian / model-based	95%	78.08%	5.1042
Fixed conformal	80%	64.00%	3.8978
Fixed conformal	95%	84.00%	7.3509
Adaptive conformal	80%	66.07%	4.1974
Adaptive conformal	95%	78.57%	6.4760

Recent-period coverage deteriorates sharply. For the Gaussian intervals, recent 15-observation coverage was 33.33% at nominal 80% and 46.67% at nominal 95%. The conformal approaches also under-cover in recent observations. This indicates that historical error calibration does not adequately represent the latest regime. Because errors overlap over multi-step horizons and the series exhibits structural change, formal conformal coverage guarantees should not be claimed.

7. Inflation surge detection

7.1 Momentum feature engineering

Ten features were constructed: 1-, 3- and 6-month changes in YoY inflation; 1- and 3-month changes in momentum (acceleration); 3- and 6-month rolling inflation means; six-month rolling standard deviation; a three-month count of consecutive increases; and a historically standardized one-month momentum Z-score. The Z-score uses a 36-month rolling reference with a 24-observation minimum and a one-month shift so the current observation does not enter its own historical reference distribution.

7.2 Objective surge definition

The surge threshold is defined as the 95th percentile of positive three-month changes in YoY inflation momentum in the training period ending December 2016. The resulting threshold is 2.3223 percentage points. A month is labelled a surge month when three-month inflation momentum exceeds this threshold.

Metric	Result
Training period	July 1993–December 2016
Positive historical 3M momentum observations	145
95th-percentile threshold	2.3223 percentage points
Detected surge months	15
Non-surge months	378
Surge-month frequency	3.82%

Detected months include August–October 2004; February–May and October 2008; October–November 2021; August–October 2025; and February–March 2026. Consecutive flagged months may belong to one episode rather than representing independent events.

7.3 Early-warning label availability

Warning window	Training positives	Test positives	Train event rate	Test event rate
Next 1 month	8	7	2.84%	6.73%
Next 2 months	11	10	3.90%	9.62%
Next 3 months	14	13	4.96%	12.50%

The low number of positive labels means a high-capacity classifier could overfit. The notebook's current evidence supports objective retrospective surge labeling and an initial early-warning data-availability assessment. It does not yet establish a reliable out-of-sample surge classifier or prove that the selected leading indicators consistently warn ahead of all episodes.

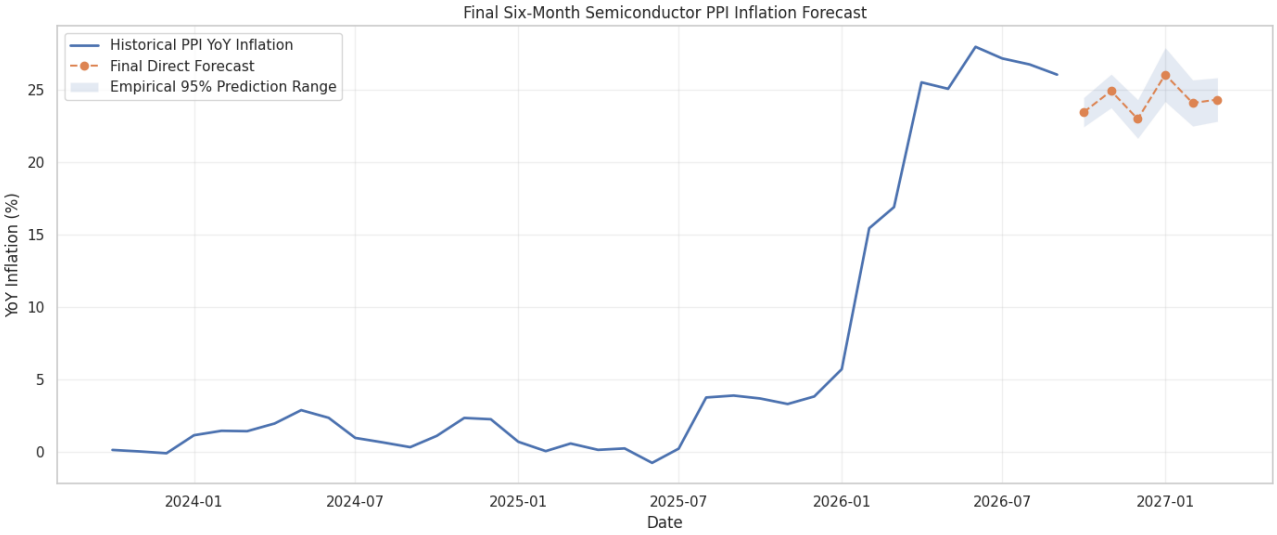


Figure 7. Final direct H1–H6 semiconductor PPI inflation forecasts with empirical 95% ranges. This is the primary final forecast chart.

8. Direct H1–H6 forecasting results

The later direct-horizon experiment evaluates persistence, Ridge, Random Forest and XGBoost separately at each horizon using 30 validation observations per horizon. The selected model is the one with the lowest validation MAE.

Horizon	Persistence MAE	Ridge MAE	RF MAE	XGB MAE	Selected
H1	0.4228	0.5349	0.3863	0.3948	Random Forest
H2	0.5453	0.7712	0.5298	0.5200	XGBoost
H3	0.7230	0.9231	0.6808	0.7379	Random Forest
H4	0.8291	1.0595	0.8764	0.8374	Persistence
H5	0.9304	1.0654	0.9007	0.8473	XGBoost
H6	1.0745	1.1174	0.9566	0.9001	XGBoost

Average validation performance across horizons: XGBoost MAE 0.7063, RMSE 0.8328, R^2 0.2092; Random Forest MAE 0.7218, RMSE 0.8442, R^2 0.1813; persistence MAE 0.7542, RMSE 0.9260, R^2 0.0101; Ridge MAE 0.9119, RMSE 1.1054, R^2 –0.4029. The average XGBoost advantage over Random Forest is modest and should not be interpreted as a statistically established difference.

8.1 Final integrated forecast from September 2026 origin

Forecast month	Horizon	Selected model	PPI YoY forecast	95% empirical interval
Oct 2026	H1	Random Forest	23.4219%	22.4093 to 24.4346
Nov 2026	H2	XGBoost	24.8866%	23.7248 to 26.0484
Dec 2026	H3	Random Forest	22.9523%	21.6148 to 24.2899
Jan 2027	H4	Persistence	26.0152%	24.1482 to 27.8823
Feb 2027	H5	XGBoost	24.0551%	22.4606 to 25.6496
Mar 2027	H6	XGBoost	24.2950%	22.7894 to 25.8005

The six forecasts are separate direct predictions, not a smooth recursive trajectory. January 2027 uses persistence and therefore carries forward the latest observed value. The empirical intervals reuse validation absolute-error quantiles; they are not guaranteed prediction coverage under structural change.

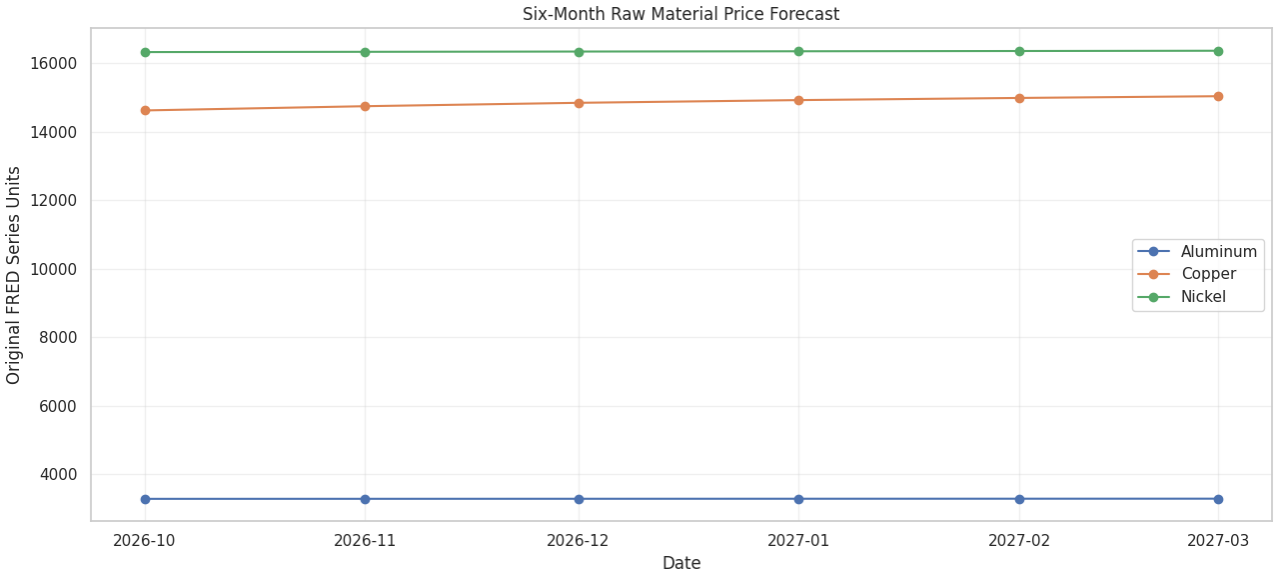


Figure 8. Six-month raw-material price projections. Include as a supplementary forecast figure; note the different original series units.

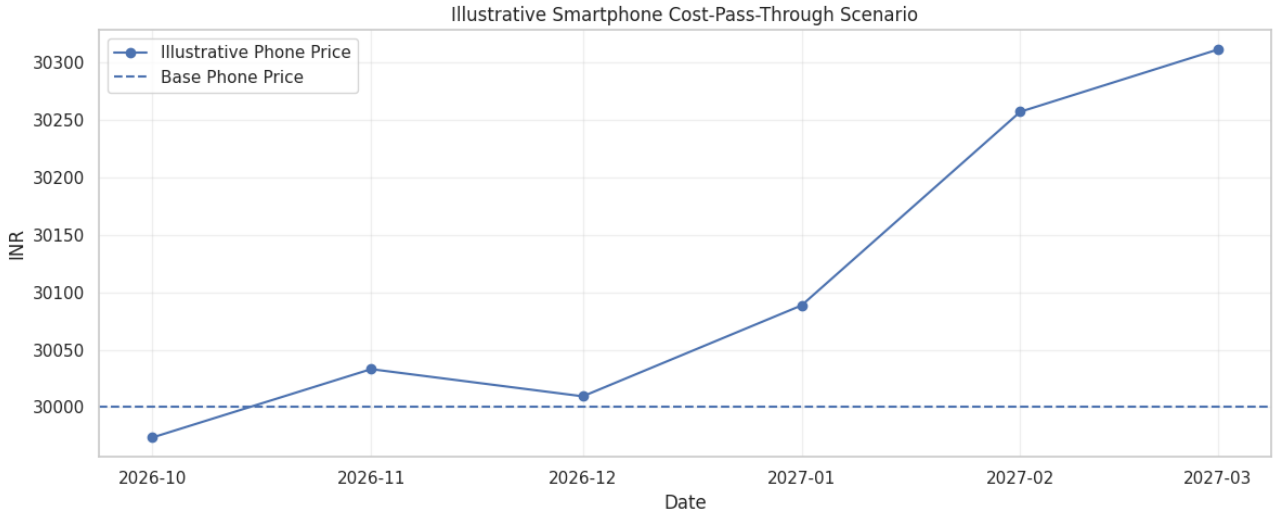


Figure 9. Illustrative smartphone price scenario. This figure is optional in the main paper and is better placed in an appendix unless the paper explicitly studies cost pass-through.

9. Raw-material and phone-cost scenario outputs

9.1 Raw-material forecasts

Copper, aluminum and nickel are forecast separately using damped-trend Holt exponential smoothing. The forecast origin is the latest available observation in the integrated run. The resulting forecasts show modest upward drift over the six-month horizon.

Material	Last observed	Mar 2027 forecast	Change from origin
Copper	14,473.056	15,043.171	+3.939%
Aluminum	3,283.054	3,289.050	+0.183%
Nickel	16,323.822	16,371.021	+0.289%

The forecasted copper series rises more noticeably than aluminum and nickel. However, these forecasts have not been accompanied by a separate rolling-origin backtest in the reported final pipeline. They should be presented as exploratory projections, not validated commodity forecasts.

9.2 Smartphone cost-pass-through scenario

The illustrative scenario assumes a base handset price of ■30,000, cost pass-through of 50%, and BOM shares of 4% copper, 3% aluminum, 2% nickel and 15% semiconductor PPI exposure. These shares are assumptions, not sourced bill-of-materials estimates. The scenario therefore covers 24% of assumed BOM exposure.

Month	Illustrative price	Adjustment vs ■30,000
Oct 2026	■29,973.66	–0.088%
Nov 2026	■30,033.10	+0.110%
Dec 2026	■30,009.47	+0.032%
Jan 2027	■30,088.49	+0.295%
Feb 2027	■30,256.80	+0.856%
Mar 2027	■30,310.88	+1.036%

Do not describe these values as predicted market prices for smartphones. Semiconductor PPI is a sector producer-price index, not a direct price series for chips purchased by a particular handset manufacturer. A publishable handset-price model requires BOM and retail-price data.

10. Discussion and research implications

The main contribution of the work is an integrated empirical workflow that compares statistical and machine-learning approaches under chronological evaluation, examines the value of manufacturing indicators, studies rolling-window adaptation, diagnoses forecast uncertainty, and introduces an objective inflation-surge label.

The historical SARIMAX experiments suggest that semiconductor industrial production can add useful predictive information relative to a univariate SARIMA baseline, especially in the moderate-inflation regime. The benefit is not uniform across regimes, and the model materially underpredicts the abrupt 2026 acceleration. This points to the limits of conventional models when the target enters a new regime.

The direct H1–H6 experiment produces low validation MAE values relative to the earlier six-month-ahead experiment, but the two experiments use different target construction, sample windows and evaluation setups. The direct results should be interpreted as a separate development experiment, not as a direct improvement over the earlier SARIMAX results.

11. Limitations

Limitation	Implication for interpretation
Structural change and out-of-distribution inflation	Models trained on earlier regimes may underpredict abrupt price acceleration.
Small number of surge labels	Only 8 positive one-month training labels; classifier claims would be unstable.
Repeated use of historical test periods	Reported comparisons are exploratory, not untouched final test results.
Overlapping multi-step errors	Standard paired tests may overstate precision; HAC comparisons are approximate.
Prediction-interval undercoverage	Historical residual distributions did not capture the recent regime.
Exogenous-variable availability	Future exogenous values require forecasts or assumptions; release lags matter.
Data vintage and revisions	FRED/BLS series can be revised; reproducible real-time tests need vintage data.
Phone BOM assumptions	Smartphone price scenario is illustrative, not a validated retail-price model.

A further real-time forecasting limitation is publication timing. A monthly observation may not be available on the first day of its reference month. A strict real-time forecasting system should align forecast origins with actual release dates and use vintage data (for example, ALFRED where available) to avoid inadvertently using revised or not-yet-published observations.

12. Recommended figures for the research paper

Figure	Recommendation	Placement
Semiconductor PPI level trend	Include	Main paper: data/EDA
Six-indicator historical trends	Include	Main paper: data/EDA
Correlation matrix	Include if space permits	Main paper or appendix
Target distribution / boxplots	Include one compact diagnostic	Appendix or EDA
SARIMAX residual diagnostics	Include	Main paper: diagnostics
Conformal / interval calibration	Include	Main paper: uncertainty
Final H1–H6 forecast plot	Must include	Main paper: final results
Raw-material forecasts	Optional	Appendix / scenario section
Phone-price scenario	Optional with explicit caveat	Appendix only
Feature importance	Appendix; not causal	Appendix

13. Conclusion

This study constructs a multi-indicator monthly forecasting framework for semiconductor and other electronic component manufacturing PPI inflation. The empirical workflow shows that model performance depends strongly on target construction, forecast horizon, estimation window and economic regime. The rolling 180-month SARIMAX with semiconductor industrial production is the strongest reported statistical specification in the earlier six-month-ahead comparison, while the direct ML experiment selects horizon-specific models for one- through six-month forecasts. The 2026 inflation acceleration exposes important limitations in point forecasting and interval calibration. The proposed surge definition provides a reproducible historical threshold, but the small number of positive events does not yet support strong claims about a deployable early-warning classifier.

The raw-material and handset-cost outputs extend the framework into exploratory scenario analysis. They should remain clearly separated from validated PPI forecasts until additional backtesting, BOM data and retail-price history are available.

14. Reproducibility checklist

Item	Required documentation
Data	FRED series IDs, download dates, observation vintages and transformation code
Target	Exact YoY PPI formula and date alignment
Features	Full feature list, lag definitions and missing-value handling
Splits	Chronological train/validation/test periods and horizon gaps
Models	Orders, window lengths, hyperparameters, random seeds and exogenous assumptions
Metrics	MAE, RMSE, R^2, bias and interval coverage definitions
Surge labels	Training-only threshold calculation and event-label horizon
Forecast outputs	Forecast origin, data cut-off date and model selected at each horizon
Code	Notebook version, package versions and final saved output tables

15. Suggested mathematical formulation

YoY inflation: $\pi_t = 100 \times (PPI_t / PPI_{t-12} - 1)$.

Direct horizon target: $y_t^{(h)} = \pi_{t+h}$, $h \in \{1,\ldots,6\}$.

Three-month momentum: $M_t^{(3)} = \pi_t - \pi_{t-3}$.

Surge threshold: $\tau = Q_{0.95}(\{M_t^{(3)} : M_t^{(3)} > 0, t \text{ in training period}\})$; $S_t = 1[M_t^{(3)} > \tau]$.

Direct forecast: $\pi\blacksquare_{t+h} = f_h(X_t)$, where f_h is selected independently for each horizon using chronological validation MAE.

Empirical interval: $[\pi\blacksquare_{t+h} - q_{\alpha,h}, \pi\blacksquare_{t+h} + q_{\alpha,h})]$, where $q_{\alpha,h}$ is the validation absolute-error quantile for horizon h. This is empirical calibration, not a distribution-free guarantee under temporal dependence or structural change.

Illustrative cost pass-through: $\Delta P_{phone}/P_{phone} \approx \rho \times \Sigma_j w_j \Delta C_j$, where ρ is the assumed pass-through rate and w_j are assumed BOM shares. This expression is a scenario assumption and should not be interpreted as an estimated economic relationship.

Appendix A. Selected notebook visualizations

The following charts are selected from the plots uploaded with the notebook. They are included because they directly support the data description, model diagnostics, uncertainty analysis and final forecast narrative. Not every exploratory plot is needed in the main paper; redundant histograms, repeated actual-vs-predicted plots and intermediate model screenshots can be omitted.

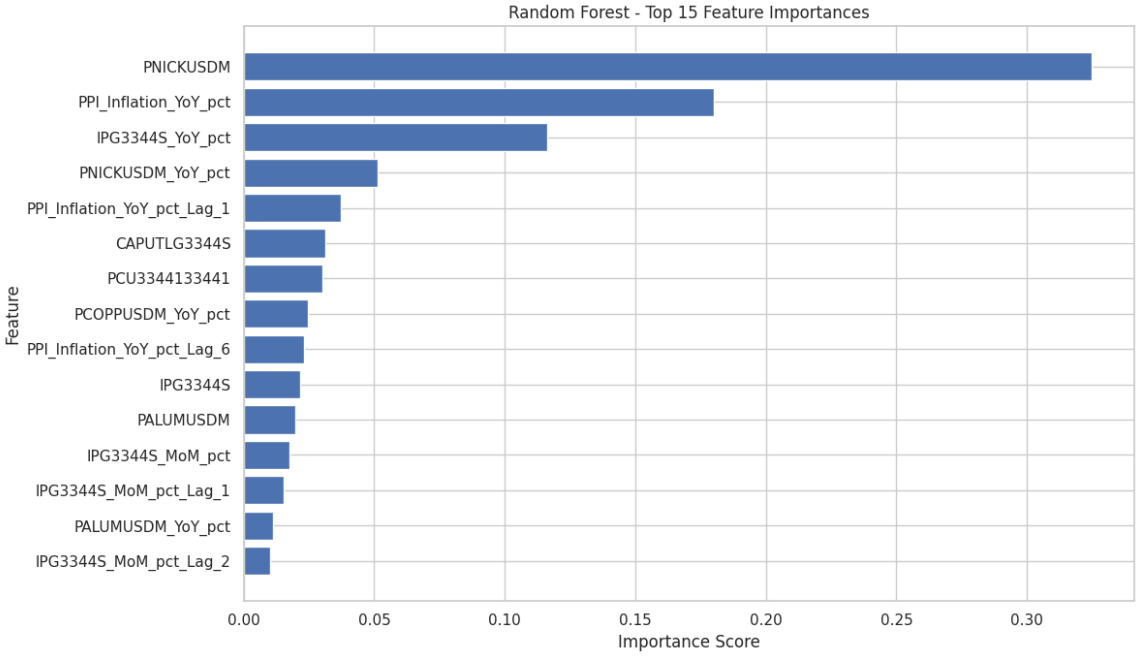


Figure A1. Feature-importance visualization. Use only as a model-specific descriptive diagnostic; impurity importance is not causal evidence.

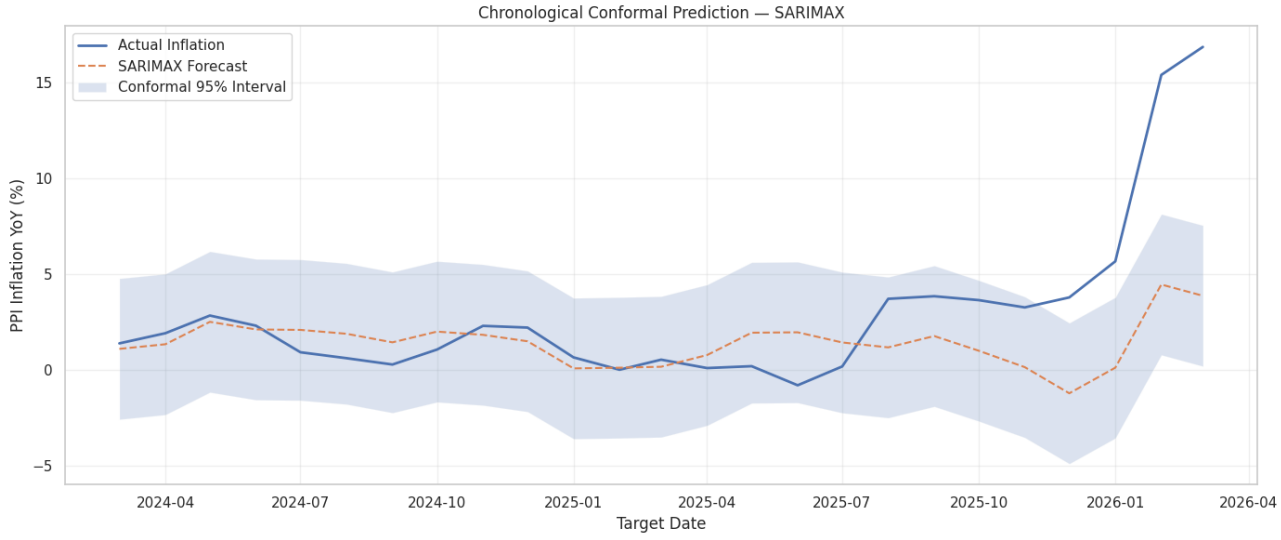


Figure A2. Historical SARIMAX forecast comparison. Useful as supplementary evidence for statistical-model behaviour.

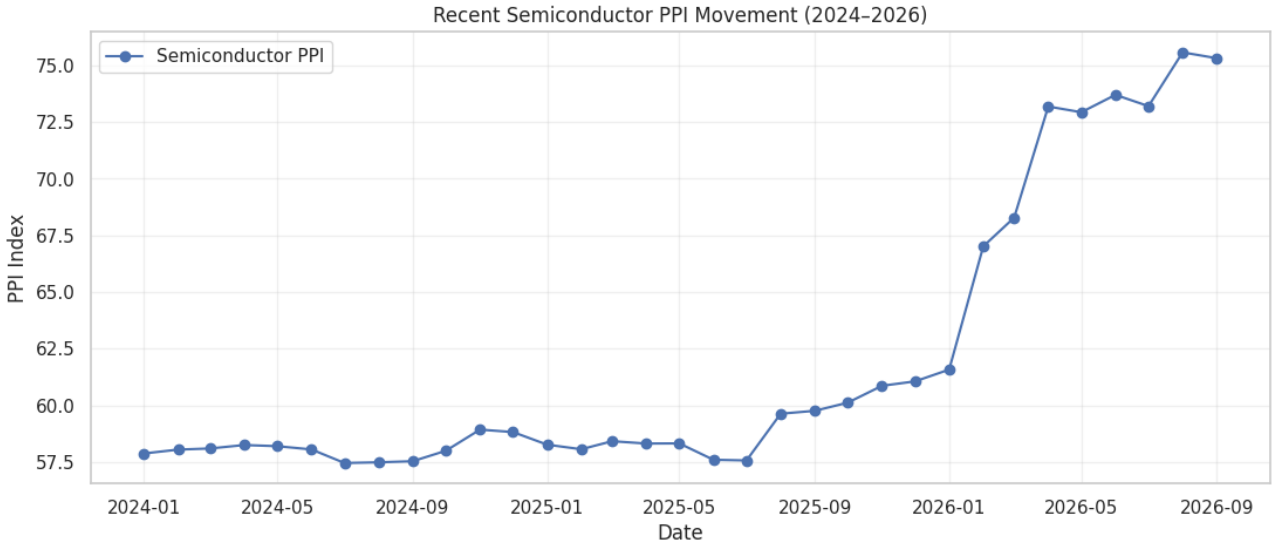


Figure A3. Semiconductor PPI level history (reproduced in appendix for convenient reference).

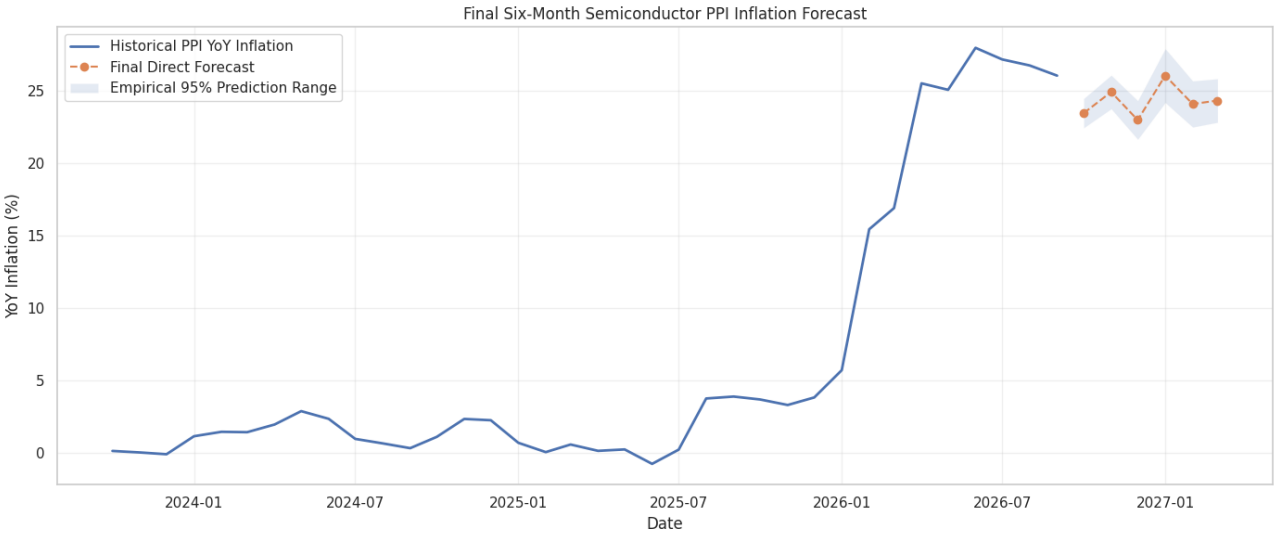


Figure A4. Final direct forecast chart (reproduced in appendix).

Appendix B. Interpretation safeguards

1. Do not claim exact replication of a proprietary Goldman Sachs pipeline.
2. Do not claim causality from correlations or feature importance.
3. Do not treat the phone scenario as a validated smartphone price forecast.
4. Do not describe empirical residual quantiles as guaranteed 95% prediction coverage.
5. Distinguish the historical six-month-ahead SARIMAX experiment from the later direct H1–H6 ML experiment.
6. State the September 2026 forecast origin and update the FRED data before making a current-date forecast claim.
7. Document data revisions and release timing for a true real-time backtest.

End of research report